INF221 - [Modern] Web Design & Development

Isaac S. Mwakabira

Objectives

By the end of this lecture, students should be able to:

- ❖ Discuss **key project roles and processes** in **software development** that are applicable/relevant to web design and development.
- ❖ Explain different **design concepts** that can be employed for **understanding** key project **requirements**, **defining** the software **project goals** and **modelling business processes** for the production of **web products**.
- ❖ Discuss different **methodologies** that can be applied in the production of web (software) products.

Indicative Content

Indicative content to achieve the above lecture objectives:

- ❖ Web development processes/activities
- ❖ Understanding the business process
- ❖ Defining the web project goals
- ❖ Composing web development team
- ❖ Process models and methodologies

Introduction to web development processes

Web development process

- ❖ Encompasses all steps, activities and methodologies involved in creating and maintaining websites or web applications

This process is constrained by availability of resources:

- ❖ Time, Money, Project scope and Technical skills.

Goals (Sommerville, 2014)

- ❖ to deliver the software to the customer at the agreed time;
- ❖ to keep overall costs within budget;
- ❖ to deliver software that meets the customer's expectations;
- ❖ to maintain a coherent and well-functioning development team

Introduction to web development processes

- It is imperative that the team carrying out a particular project breaks down the problem in manageable chunks/activities/tasks,
  - so that they can be worked on methodically - using particular models and methods.
- The business organization usually identifies a PRODUCT owner or PRODUCT manager to
  - identify technical skills required in the project
  - monitor and report progress to the project stakeholders
  - manage resources allocated to the project
  - facilitate communication among developers and any other stakeholders

Web/Software Processes

- ❖ Process
  - ➢ A software process is a set of related activities that leads to the production of a software system.
- ❖ Process Model
  - ➢ A software process model (sometimes called a Software Development Life Cycle or SDLC model) is a simplified representation of a software process.
  - ➢ Each process model represents a process from a particular perspective and thus, only provides partial information about that process ie. Waterfall, Spiral, V-Shaped, Incremental development
- ❖ Methodology/Software engineering method
  - ➢ Structured approach to software development which includes system models, notations, rules, design advice and process guidance. Ie Rational Unified Process, Scrum, XP, Pair Progmng.

This presentation gives a general overview of software process models and methodologies that may be used in the production of any software product

- ❖ web or desktop or mobile based (Details will be covered in 3rd year)

Key Processes

Key processes activities are:

- ❖ Planning and Design, which create the appearance, layout and style that users see
- ❖ Development and Testing, which brings this imagery to life as a functioning web tool

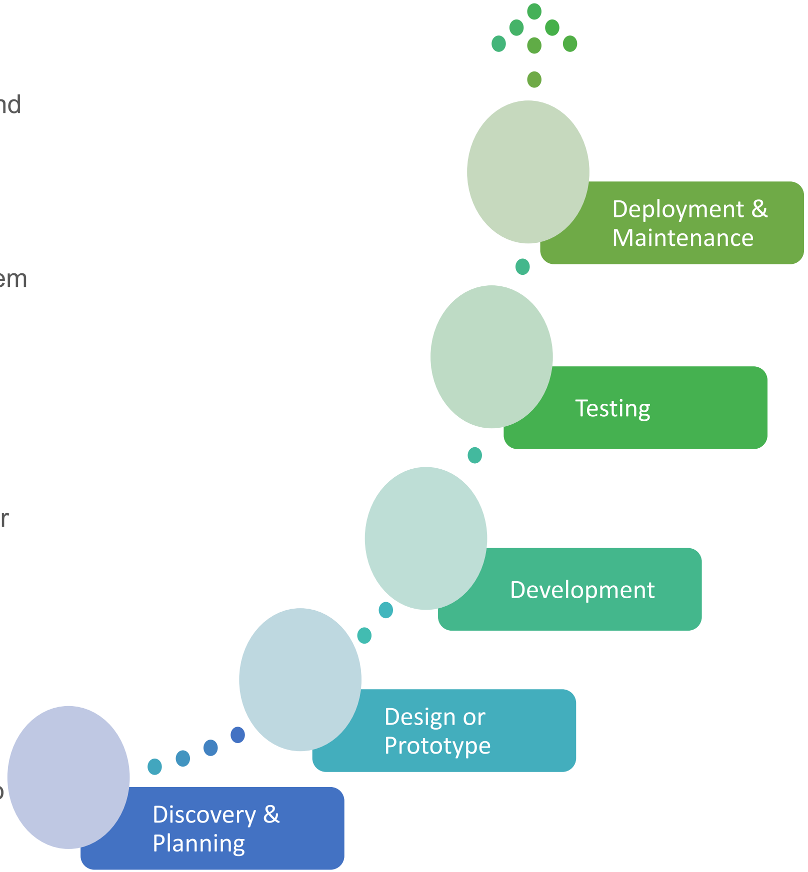
Iteration is key throughout these core activities

To manage change and risk of failure, the systems life cycle breaks a project down into a series of well-defined stages

- ❖ from the formulation of the initial concept through to the implementation and maintenance of a fully working system

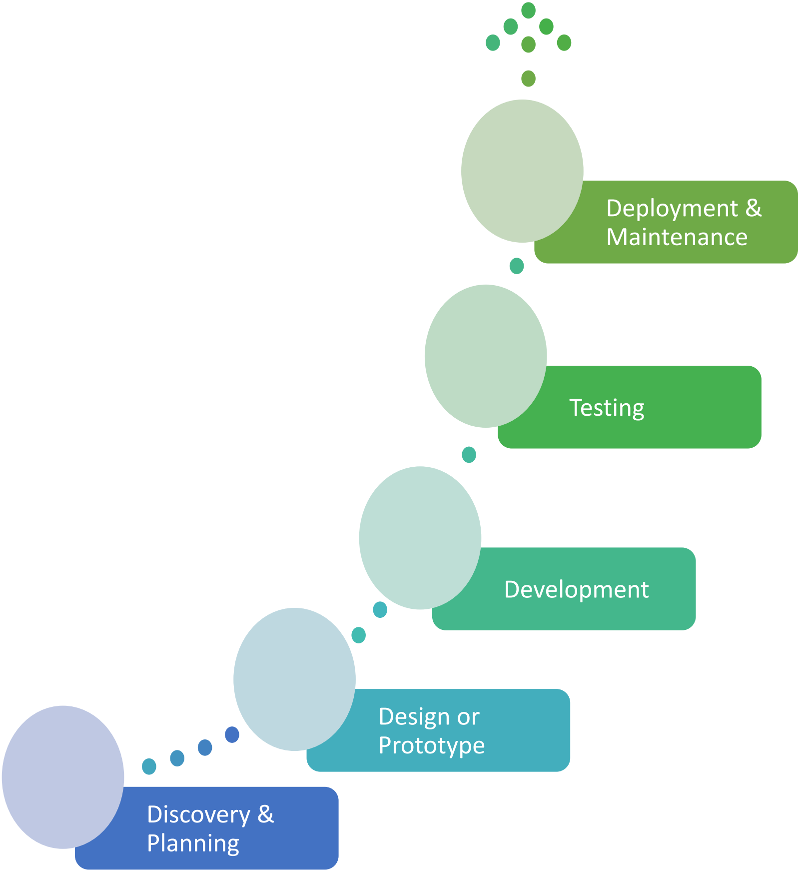
Project Life Cycle Stages

- Planning and Discovery
  - Problem definition, team composition, skill-set required, software and hardware development requirements,
- Requirements analysis or specification
  - the scope of the system is defined, and a set of requirements is derived and defined
    - i.e. the functionality and levels of service that the new system must provide
- Design, Development & Testing
  - designed to meet the requirements specified in the requirements specification
  - software product is built - database and software applications are written, and data is loaded into the system
  - depending on the model in use, testing may be done during, or after actual development & deployment of the system software
- Deployment, Maintenance & Support
  - once the client has accepted the new system and it is in daily use
  - there will inevitably be adjustments to be made to the system throughout its useful life
  - hardware elements will need to be replaced or upgraded
  - and the need for additional functionality
  - or information processing features may necessitate modifications to software or database programs



Web Dev. Processes - **Planning** & Discovery

- Planning for proper software management
  - People Management - Team composition
  - Defining project goals & scope
- Why manage projects?
  - Organisational budget
  - Time - schedule constraints
- Group Working
  - Each member’s duty is to:
    - Understand key issues that influence team working
      - such as team composition, organization, and communication
      - the same can be reflected in the collaboration tool being used



Project Management - **Team** Composition

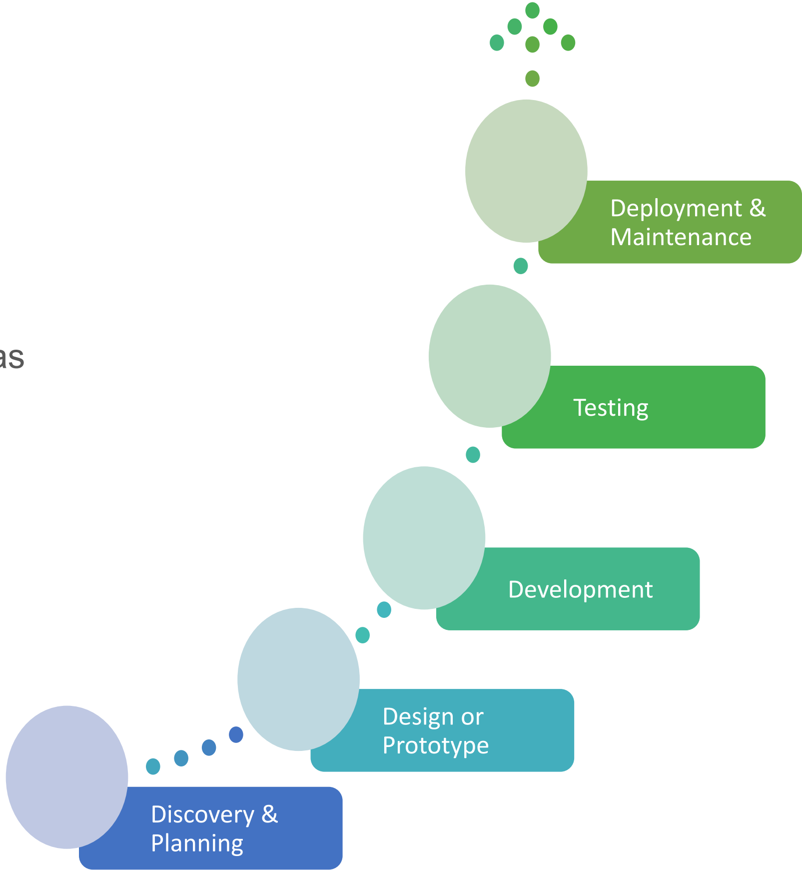
- Project success is based on choosing the right people for the project team and obtaining the necessary level of commitment from them
- The list below can be used as a guide to putting together a balanced team:
  - A single coordinator or shaper (but not both) as team leader
  - A planner to generate ideas
  - A monitor-evaluator to provide objective analyses
  - One or more:
    - implementer
    - team worker
    - resource investigator
    - completer-finisher
- Task allocation
  - The work breakdown structure (WBS) can be used to identify the work to be done
    - after which you will have a reasonable idea of the work activities that need to be carried out,
    - the number of staff needed to undertake the work
    - and the specialist skills and knowledge required

Project Management - Defining **Project Goals & Scope**

- Planning a project is about deciding what work needs to be done in order to achieve the objectives
  - who will undertake each task involved
  - when each task will be carried out and how long the work will take
  - and what resources will be required.
- In addition, all of these activities must be viewed against the backdrop of cost
  - both ongoing and final.
- The plan addresses questions such as ***what***, ***when***, ***how*** and ***by whom***
- User requirements define the project deliverables
  - which determines what needs to be developed
- Project scope defines project boundaries
  - determine both the extent of the resources required
  - and the amount of time needed to complete the project

Web Dev. Processes - Planning & **Discovery**

- Discovery – Requirements gathering
  - Users must be involved – they are critical
  - Input
    - Reports from clients and documentations from Business Analyst
  - Output
    - Complete final project documentation with requirement specifications for designers as well as developers
      - wireframes, prototypes
- Discovery Process
  - Involves the understanding and application of essential design concepts for defining the problem and proper design solutions
    - Problem statement, User Stories, Personas, Scenarios, Use Cases
  - Key is to understand how **information flows** within the **business processes** (design problem)
    - how it is generated, processed and stored



Discovery & **Modelling** of *Business Processes*

To discover and model business requirements, certain concepts maybe used:

- ❖ User Stories
- ❖ Personas
- ❖ Scenarios
- ❖ Use-cases

It is imperative that **YOU** know and understand why each concept is important and how it can be used in modelling business processes.

Discovery & **Modelling** of *Business Processes*

- User story
  - Is an informal, general explanation, using natural language, of a **software feature** written from the perspective of the end user or customer.
  - Describes how a particular **piece of functionality/feature** will deliver business value to its user.
- Persona
  - Means a **fictional** character **depicting** a particular user who will use the software system.
  - Used in the initial stages of software development to know **characteristics and capabilities** of system users **while** designing and developing a particular system
    - users' needs, experiences, behaviors, and goals

Discovery & **Modelling** of *Business Processes*

- Scenario
  - Is a scene that illustrates some interaction with a proposed system
  - Is a tool used during requirements analysis to describe a specific use of a proposed system.
  - Scenarios capture the system, as viewed from the outside, e.g., by a user, using specific examples

Scenarios are useful in discussing a proposed system with a client, but requirements need to be made more precise before a system is fully understood

Use-cases come the rescue.

Discovery & **Modelling** of *Business Processes*

To discover and model business requirements, certain concepts maybe used:

- Use cases
  - Use cases are a way of describing interactions between users and a system using a graphical model and structured text
  - They model actual functional features of the resulting system.
  - Can be graphically or textually represented
    - At this level we will stick to textual representation

- Examples (Student's Library Access System):
  - Check-in the library, includes:
    - Retain student's restricted items at the reception
  - Charge student for lost book (s)
  - Charge student for overdue book (s)
  - Borrow a book or books (s)
  - Return borrowed book (s), includes:
    - Pay for overdue books (s),
    - Pay for lost book (s)
  - Reserve reading space, includes:
    - Pay for reserved space
  - Reserve book (s) at the special counter
  - Check-out from the library, includes:
    - Return student's retained items

Web Dev. Processes - **Design**

Understanding & Conceptualizing Design

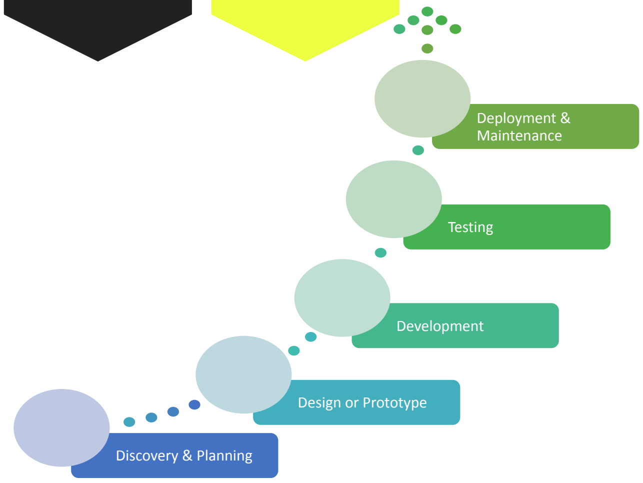
❖

It includes 4 key steps:

❖ Establishing requirements
❖ Developing alternatives
❖ Prototyping
❖ Evaluating

Design process

❖ **Information Architecture**
  ➢ How should we structure pieces of information on a web page?
  ➢ Wireframing, Prototyping
❖ May also include the use of: storyboards etc

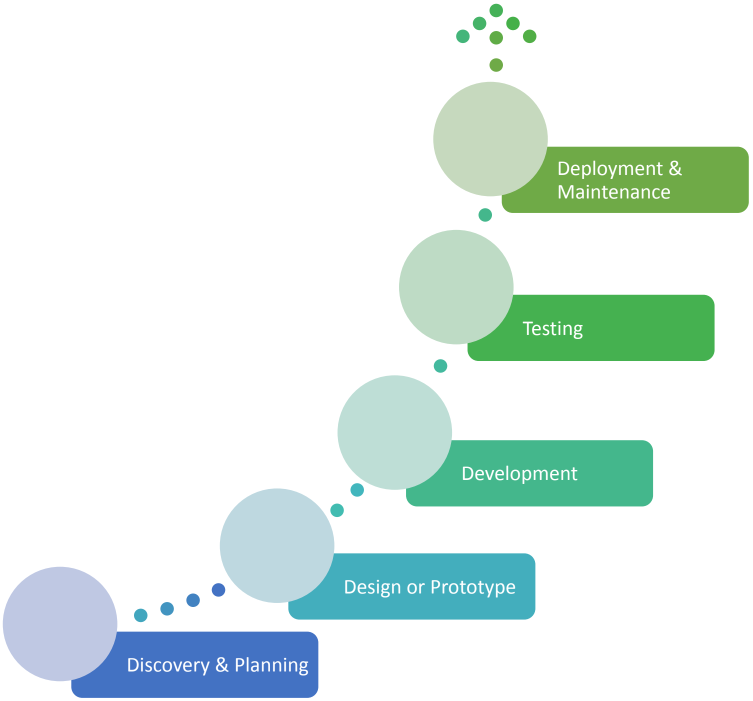


Design - Business Processes & Users

- The key principle to good design is:
  - To understand your users - they are the people who will actually be using and interacting with your web application.
    - Who are they?
    - What activities are being carried out?
    - What are they looking for?
    - What are their objectives?
- Providing quality user experience (UX) must be central to the process.
  - Need to optimize the interactions users have with a product:
    - So that they match the users' activities and needs
  - Understanding users' needs
    - Take into account what people are good and bad at
    - Consider what might help people in the way they currently do things
    - Think through what might provide quality user experiences
    - Listen to what people want and get them involved
    - Use tried and tested user-centered methods

Web Dev. Processes - **Development**

- Development
  -
- Tools for managing web development activities
  - Collaboration
  - Development IDEs
  - Testing Frameworks



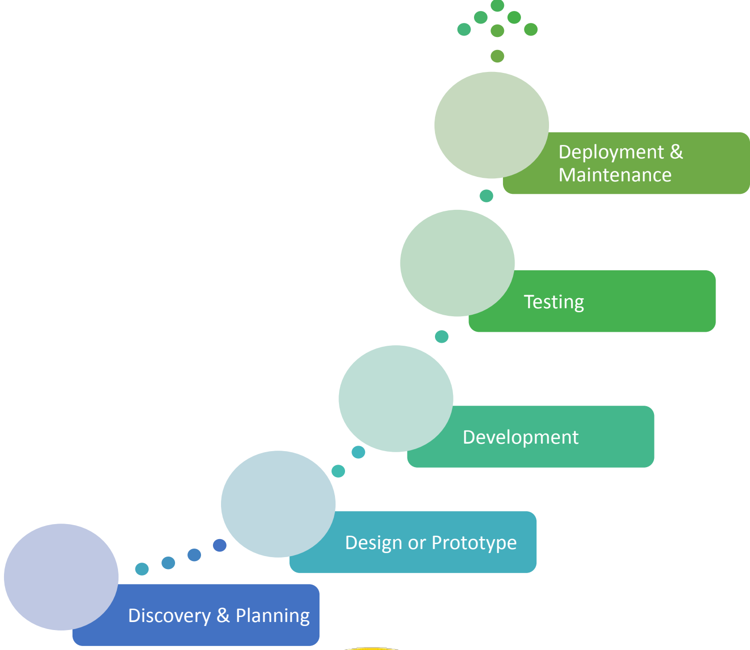
Web Dev. Processes - **Testing**

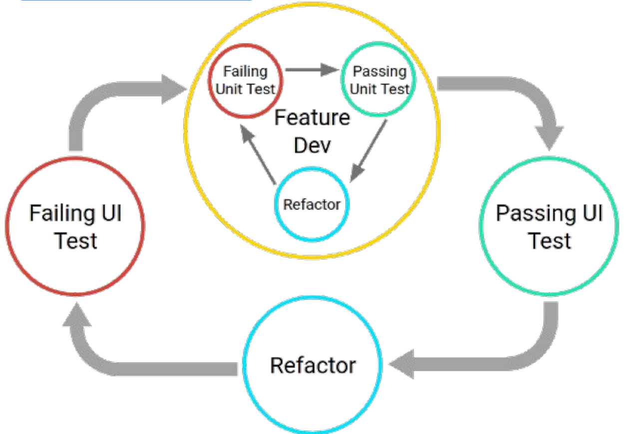
Testing

- ❖ Unit Testing
- ❖ UI Testing
- ❖ Integration Testing
- ❖ E2E

Iteration is key throughout testing process

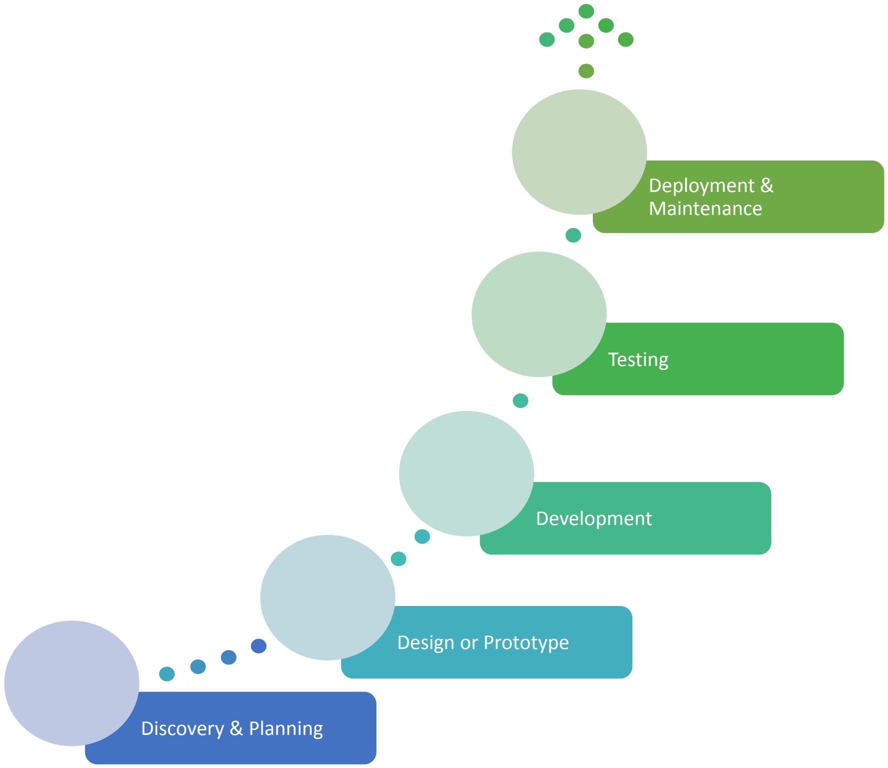
- ❖ usually done together with design and development





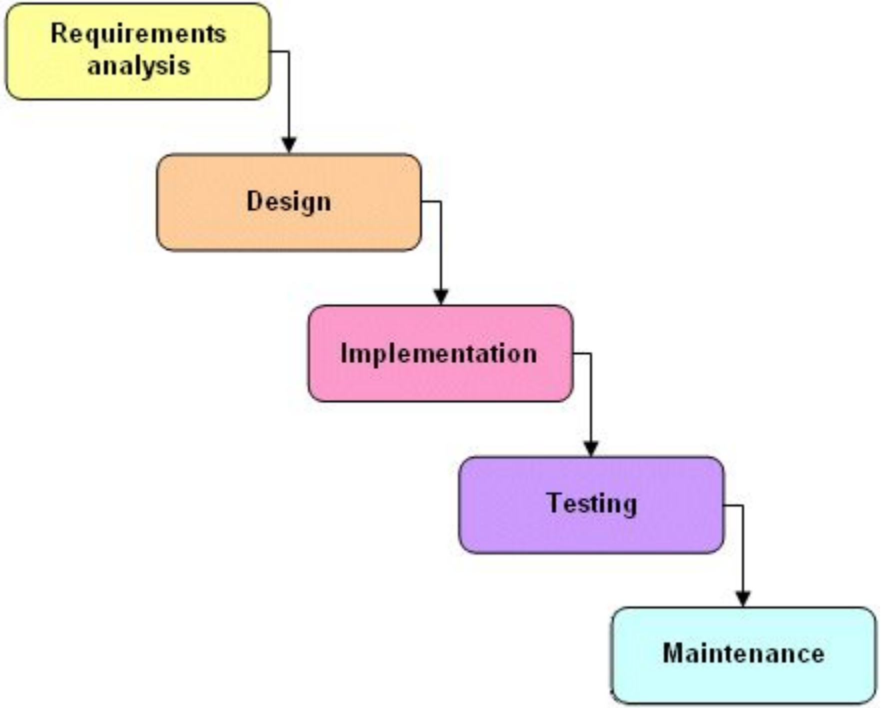
Web Dev. Processes

- Deployment – production
- Finished product
  - enables users to achieve their daily work
- Maintenance – mostly forgotten activity
  - especially in most projects at home – Malawi.

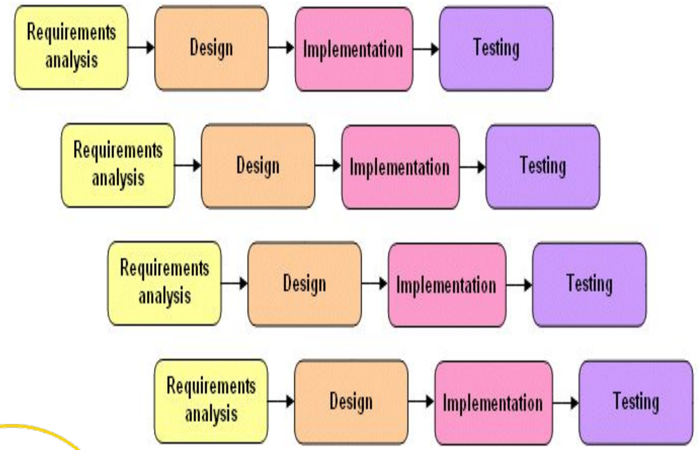


Development Lifecycle Models - Waterfall

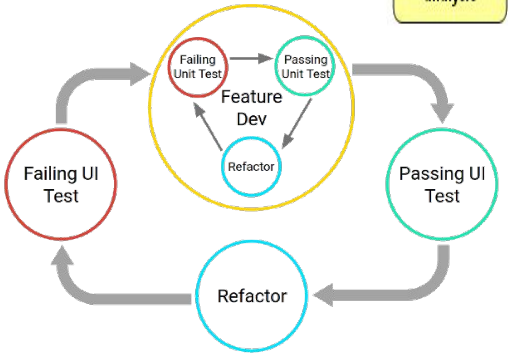
- move to the next phase only when the current phase is complete
- proceed from one phase to the next in a purely sequential manner
- phases of development are discrete, and do not allow for jumping back and forth, or any overlap between the phases
  - although there are modified versions of the waterfall model that are not so rigid



Development Lifecycle Models - Incremental Model

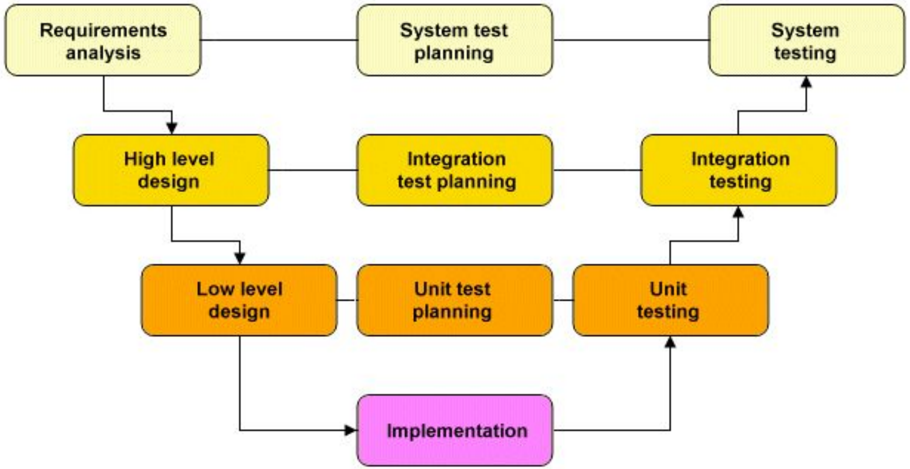


- Allows different parts of the system to be developed separately, each with its own life cycle iteration
- Each iteration includes a requirements analysis, design, implementation and testing phase
  - The first iteration usually produces a working version of the system on which subsequent iterations build.
- The incremental model provides a more flexible and less costly alternative to the waterfall model
  - in which changes to the scope or requirements of the project can be more easily accommodated.
- Each iteration can be completed relatively quickly, and is easier to manage.
- Like the waterfall model, however, each phase of each iteration must be completed before the next phase can commence,
  - and the overall system architecture may be poorly defined in the early stages of the project and evolve it as project progresses



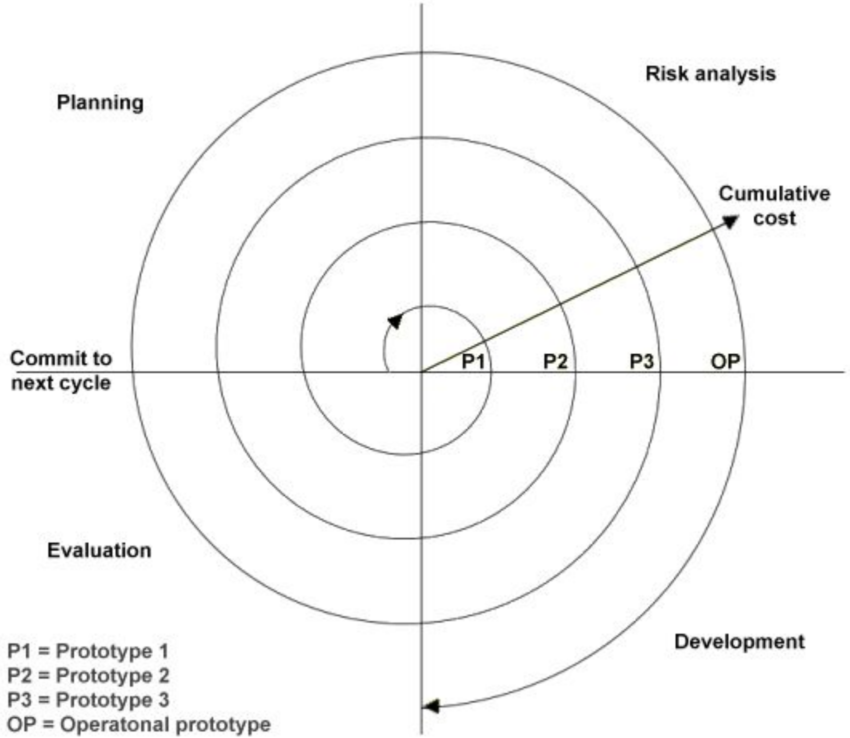
Development Lifecycle Models - The V-shaped model

- Like the waterfall model, each phase of the V-Shaped life cycle must be completed before the next phase begins
- Much more emphasis on testing
  - Test procedures are developed early in the life cycle
    - unit testing is planned to assess the success of the low-level design phase, which concentrates on individual system components
    - integration testing is developed during the high-level design phase, which focuses on the overall system architecture
    - system testing is designed to assess whether the requirements determined during the requirements analysis phase have been met



Development Lifecycle Models - The Spiral model

- The spiral model is similar to the incremental model
  - with more emphasis on risk analysis
- Project passes repeatedly through the four phases of planning, risk analysis, development and evaluation
- Cost of the project increases as the spiral goes through successive iterations
  - high degree of risk analysis and the production of a prototype system early in the life cycle makes this a good option for large and complex projects
    - although it can be potentially costly, and
    - the level of expertise required for the critical risk analysis phase means that it is not really suited to smaller projects



Development Methodologies

Describe a framework for the development of information systems

- ❖ A particular methodology is usually associated with a specific set of tools, models and methods
  - ➢ for the analysis, design and implementation of information systems
  - ➢ and each methodology tends to favour a particular lifecycle model

Development Methodologies - Agile Methodologies

- ❖ Refers to a group of loosely related software development methodologies that are based on similar principles [**any examples???**]
- ❖ Each iteration of the project life cycle results in the production of working software, which is then evaluated by the customer before the next iteration begins
- ❖ Production of working software, rather than the completion of extensive project documentation, is seen as the primary measure of progress
  - ➢ The software produced at the end of an iteration has been fully developed and tested, but embodies only a subset of the functionality planned for the project as a whole.
    - ■ The aim is to deliver functionality incrementally as the project progresses.
    - ■ Further functionality will be added, and existing features will be refined, throughout the life of the project

Question Time

- List and explain any advantages and disadvantages of each model discussed above
- List & explain any advantages and disadvantages of each methodology

Summary

Key Processes

- ❖ Understanding the business process
  - ➢ Defining the web project(s) goal(s)
- ❖ Web development team
- ❖ Web development phases

Development Models & Methodologies

- ❖ Waterfall, Spiral, V-Shaped
- ❖ Agile Methods - Scrum, Extreme,

References

- List of software development philosophies - Wikipedia accessed on 10/03/2024
- Sommerville I. (Sept. 2014). Software Engineering, 10th Edition
- Website development team structure: roles and processes accessed on 10/03/2024
- Waterfall, Agile, Kanban, & Scrum: What's the Difference? [2024] • Asana accessed on 10/03/2024
- What Is A Technology Stack? Tech Stacks Explained | MongoDB accessed on 10/03/2024